Event Menu Recommendation via Sentiment Analysis
Empire's Taste

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INTRODUCTION

Designing menus for events in New York City remains largely intuition-driven, despite the availability of rich culinary media and structured recipe databases that could support more informed decisions. Prior work has applied sentiment analysis to food reviews [1] and proposed context-aware recommendation systems [3], but these solutions tackle each problem in isolation. The main challenge is integrating unstructured public food opinion with structured recipe data into a single, automated pipeline that translates daily media coverage into actionable menu guidance.

GOAL

Can culinary media be systematically analyzed to identify trending food topics and drive data-informed recipe recommendations for specific events?

The outcome is Empire's Taste, an automated pipeline that daily harvests culinary news, computes sentiment and trend indicators, and delivers actionable menu insights through interactive dashboards.

PROPOSED SOLUTION

Empire's Taste collects daily culinary news from Eater NY, restaurant reviews via web scraping from OpenTable, and queries Spoonacular using food trends extracted via TF-IDF and spaCy, processing data through a three-layer Medallion architecture — Bronze, Silver, and Gold (Figure 1) — orchestrated with Apache Airflow, applying VADER sentiment analysis and PySpark aggregations, and delivering results through Plotly Dash dashboards and a Next.js web application for event planners.

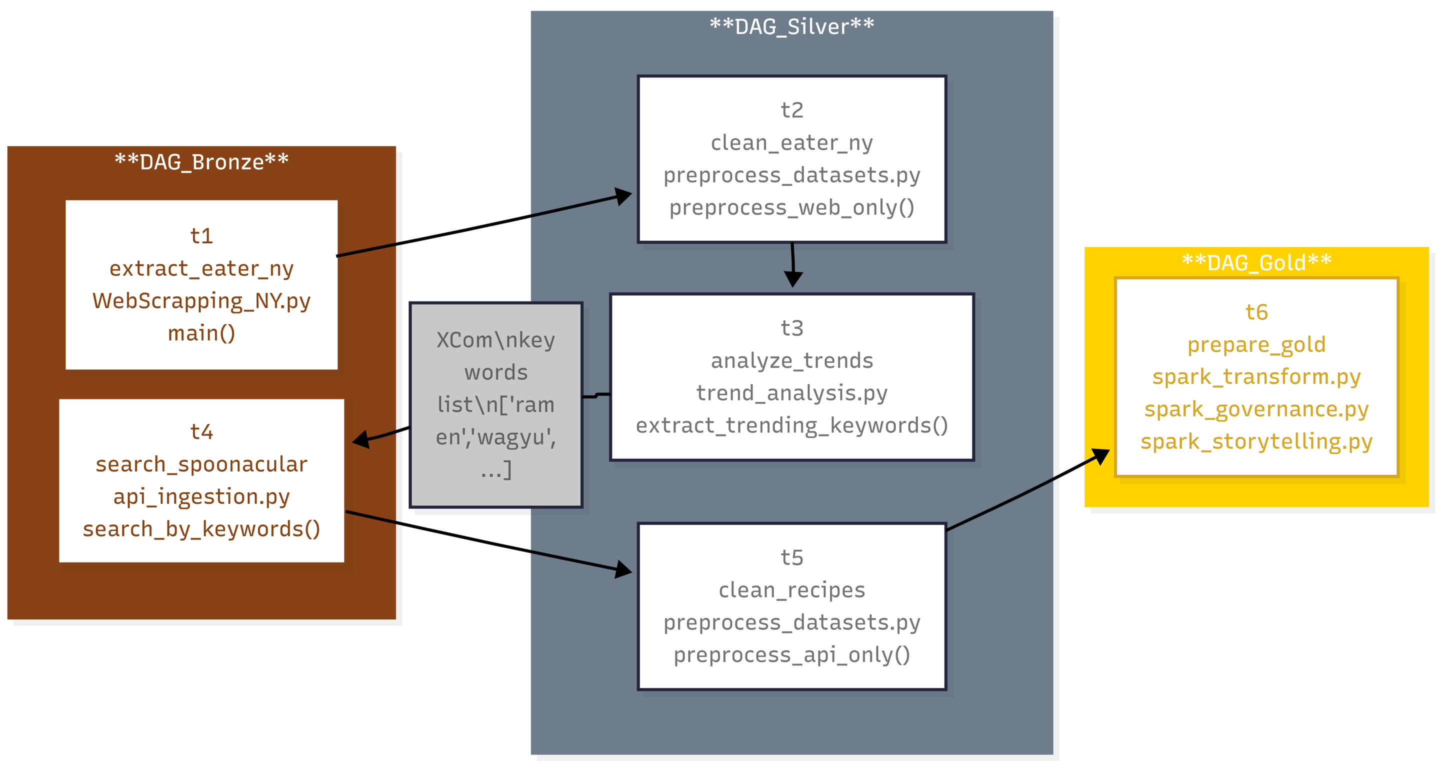
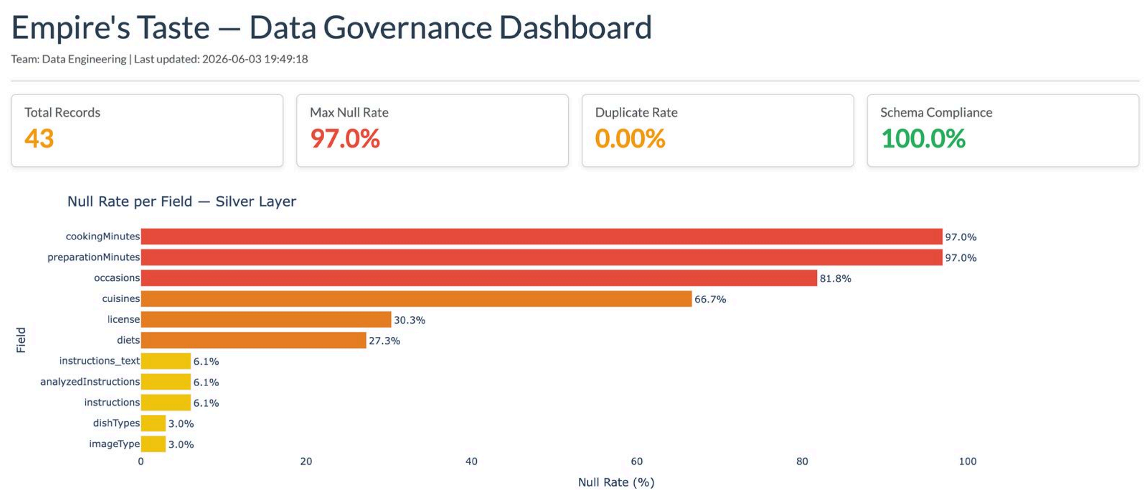
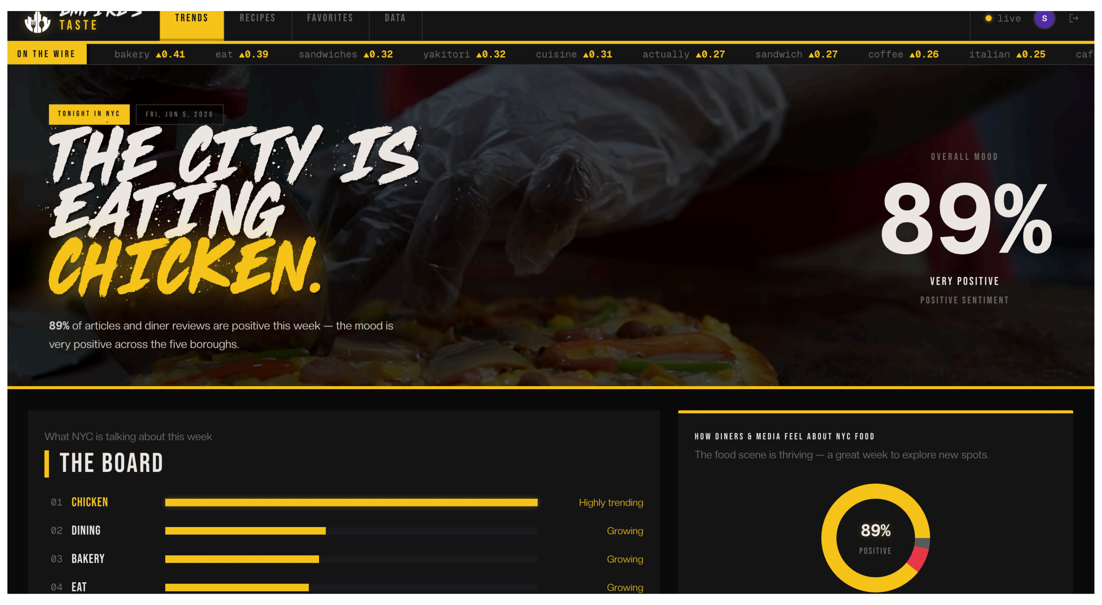


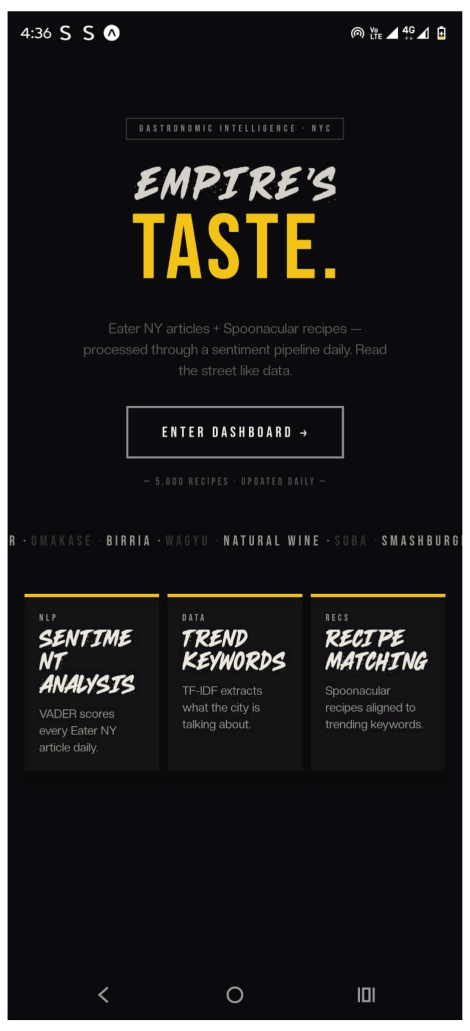
Figure 1. Medallion architecture

EXPERIMENTS

The hybrid TF-IDF and spaCy approach was tested against naive frequency counting. While naive counting returned terms like security and terminal, the proposed method correctly identified food-relevant keywords like pizza and sushi to query Spoonacular. Data governance KPIs confirmed 100% schema compliance, 60% NLP token reduction, and correct identification of null rates in Spoonacular fields as expected API behavior.







RESULTS

Empire's Taste successfully combines sentiment analysis, trend extraction, and recipe recommendation in a single automated pipeline, surpassing prior solutions that addressed these problems in isolation.

Feature	Hu & Liu [1]	Adomavicius [3]	Empire's Taste
Sentiment analysis	✓	✗	✓
Recipe recommendation	✗	✓	✓
Real-time trend extraction	✗	✗	✓
Event-specific context	✗	✓	✓
Automated pipeline	✗	✗	✓
Data governance	✗	✗	✓

Tabla 1: Comparison with previous solutions

CONCLUSIONS

Empire's Taste enables event planners to identify trending dishes in NYC, understand public sentiment [2] toward specific foods, and receive tailored recipe recommendations, demonstrating that culinary media can be systematically transformed into actionable, data-driven menu guidance.

BIBLIOGRAPHY

[1] Hu & Liu, "Mining and summarizing customer reviews," ACM SIGKDD, 2004.
[2] Hutto & Gilbert, "VADER: Sentiment Analysis of Social Media Text," ICWSM, 2014.
[3] Adomavicius & Tuzhilin, "Context-aware recommender systems," Springer, 2011.